Tragedy of the Commons

Facilitation Materials – Discussion Guide

ToC is a very versatile game suitable for a number of different contexts. It is up to the facilitator to determine how they want to use ToC, for what purpose and in what sort of class environment. This **Discussion Guide** will provide information about how to adapt the game to fit those educational needs and how to foster learning within players through discussion.

This guide is intended for teachers who have already read the Introductory Guide.

How to Incorporate ToC in the Classroom

Preparing Students Beforehand	• Facilitators should prepare their students for a "different day" before they play the game. ○ Tell them they will be using their mobile devices for a classroom activity. ○ Get them excited about the game-like lesson, but give clear and strict guidelines for use of devices in the classroom.
How a Typical Game Runs	• The general process is that the class will: 1. play through the game once until all fish (mostly) disappear. 2. have a discussion to come up hypotheses for why the fish disappeared and how to prevent it 3. replay the game once or more to test these hypotheses. ○ These steps can happen over as long of a time period as is convenient for the facilitator and class. • Typical game length: ○ In general, playing a few seasons of the game takes 20-30 minutes. ■ This is flexible, though, since each game of ToC can have as few or as many seasons as desired, and seasons can be as long or short as desired. ○ Defining how long seasons should be is up to the facilitator. ■ For example, seasons can be time boxed such that they can only be five minutes long, or seasons could continue until all or most players have run out of fuel, or seasons could end when most students have caught a certain amount of fish. ■ It is completely up to the facilitator, and it is not necessary to tell students exactly how seasons are defined.

Suitable Settings for Use	ToC is designed: ● for a wide age group. ○ It can be used in elementary, middle, and high school classes, as well as even higher education settings. ● to work with many different subjects. ○ It can be used in science classes, such as ecology or biology, or social science classes like economics, or anywhere else. ● to fit into classtime in many different ways. ○ It can be run at the beginning of a unit as an introduction to its concepts, or at the end of a unit as a review of those concepts and an opportunity for students to prove what they learned, or both. ○ It can be run every day for a period of time, or once a week, or with any other frequency. **Facilitators can (and should!) be creative about how they incorporate the game and adapt it to fit their needs.**

Steps for Facilitating Discussion

During the First Playthrough	● As a facilitator, the two most important things to keep in mind are: ○ Give players as little information as possible for them to start playing the game. ■ Explain that it's normal to be confused about what's happening in the game. ■ Avoid answering questions. Save them for the discussion after the first playthrough of the game. ■ Tell players their objective is to be a "successful fisher." It is up to them to determine what that means! ■ **It is imperative to not tell players what to do.** ○ Encourage players to make observations, come up with questions, and listen to other players. ■ Ask players what they see on their screens. ■ Make sure players pay attention to what other players are saying. ■ If players ask questions, say "Good question!" and tell them that they'll have a chance to ask each other questions during the pauses between seasons.
During the Discussion	● The job of the facilitator in this part of the game is to initiate a student-driven discussion while pointing out concepts and questions that are important for players to explore in the next playthrough(s). ● To do so, a facilitator should: ○ Record notes on a chalk/dry-erase board. ■ Try organizing notes as observations, questions, guesses, ideas, proposed experiments, etc.

- Name the target concepts and key terms as players describe them.
  - See the next section, **Key Concepts**, for more details on this.
- Embrace all players' ideas, even those that are off track, and gently guide players to a consensus about how the game will be played in the next playthrough.

- The following table outlines how a discussion shifts from observations about each individual's experience, to a process of analyzing players' collective experience, to a proposed experiment that will answer players' questions:

Common Facilitator Questions	Common Player Responses
Observation: What happened? What stations did you visit? What happened to you? What did you do in the game? What were your strategies? What did you do when you noticed the number of fish decreasing? What questions do you have?	I fished, and sometimes it was more productive, sometimes it was less so. As time went on, I started getting more fish because I bought better materials. I sat down during the first few seasons to save fish for later. There were fewer fish so I started only fishing at ___ stations/stations with fewer people. Why did the number of fish go down? Can the population decline be prevented?
Hypothesis: Why do you think the number of fish decreased? How? How did it start? Did you notice any pattern? Was anything not affected? Develop a theory:	Stations had less fish when many players fished from them. Stations that were…
Experiment Design: Now you have a chance to play the game again, with the exact same starting conditions. How will you play differently in order to answer your questions and test your ideas? How might you keep the most fish in the game for longer?	Let's have only a certain number of people go to each station. Let's limit the number of fish a player can have in each season. Let's have players with more fish donate/trade to struggling players.

After the Discussion	• By the end of the discussion, students should have developed ideas for how to experiment with strategy in order to prevent the depletion of the fish population. • These experiments should be encouraged and noted. • Replaying the game one or more times gives players the opportunity to test the new strategies with each other.

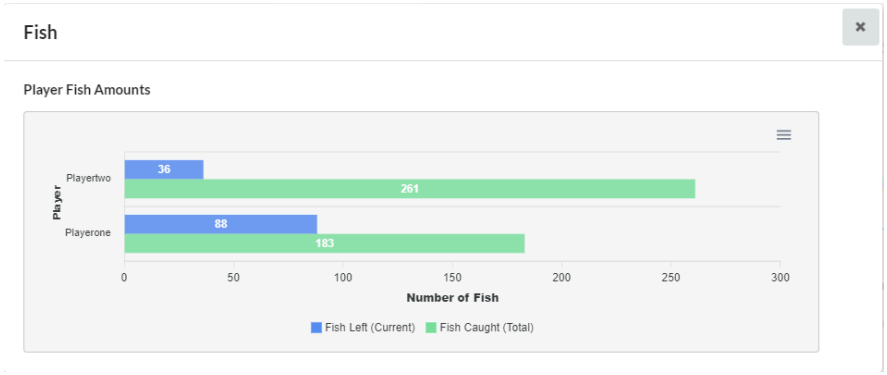
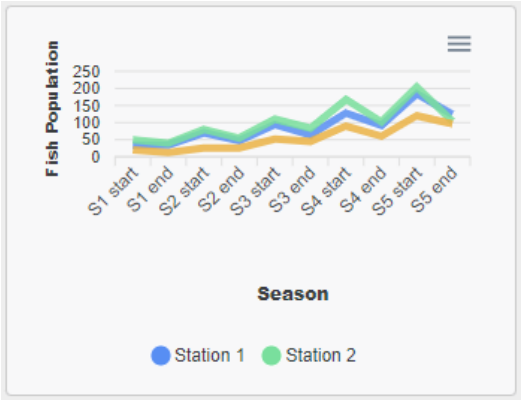
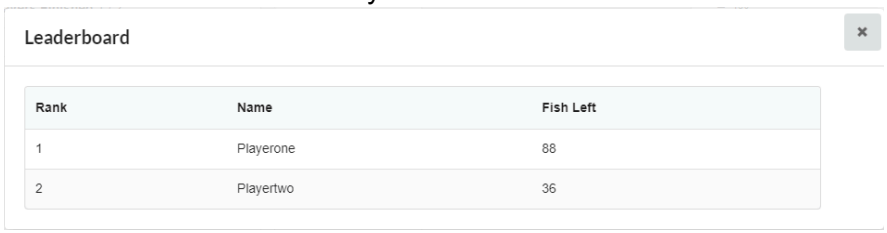
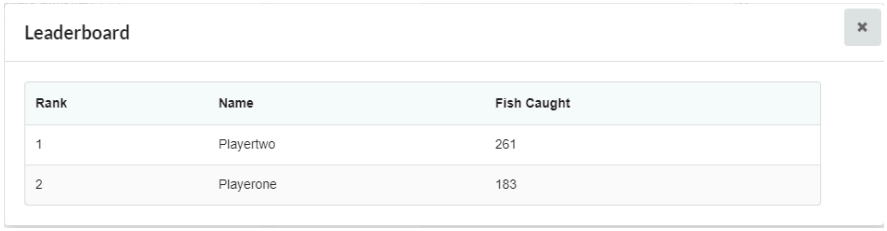
Key Concepts

During the discussion, players will describe many of the concepts in the following lists without necessarily using the exact terms listed here. The job of the facilitator is to give names to these core learning goals as players "discover" them.

For example, when a player suggests, "We fished at some stations so often they were not able to recover or produce more fish," it might be appropriate for a facilitator to respond, "So, you're wondering if there's a maximum sustainable yield?" and write the term on the board.

Science Concepts	• Population size ○ The number of individual organisms within a population (BioLibreText). • Initial population ○ In the case of ToC, this is each station's starting number of fish that the facilitator determined during setup of the game. • Population growth rate ○ The rate at which the size of a population is changing over time (Nature). • Harvest rate ○ the proportion of a population that is killed by humans for economic purposes (in ToC, this would be fish caught). For a specified population and time period, it equals number harvested over that time period divided by initial population (Biology Stack Exchange). • Maximum sustainable yield (MSY) ○ The maximum yield (or catch) that can be taken from a species' population over an indefinite period while still allowing the population to sustain itself at a healthy level (Science Direct). • Equilibrium ○ In an unexploited state, when a population is on average in balance with its environment such that birth and growth are offset by death and the population is not increasing or decreasing. (Science Direct). ○ The fish populations represented in ToC have no equilibrium because there is no natural death. Instead, all the population decline comes from the exploitation of the players' fishing. This allows the MSY to be higher and the game to last longer. • Quota ○ A fixed minimum or maximum number of individuals that are permitted to meet some criteria ○ This concept could be involved in players' strategies, for example

	setting a quota on how many fish each player is allowed to catch, or a quota on how many players can fish at each station. ● Overfishing ○ The removal of a species of fish from a body of water at a greater rate than that the species can replenish its population naturally, resulting in the species becoming increasingly underpopulated in that area. ● Exponential rate ○ A per capita rate of growth that takes the same positive value regardless of population size, due to the condition of unlimited resources and no predation (BioLibreText). ○ In ToC, the station fish populations exhibit exponential growth since they always double after each season. While this is not accurate to real world fish populations, it helps demonstrate the game's point about overfishing, as even with exponential growth, the station populations still decrease by large amounts due to players' fishing.
Economics Concepts	● Tragedy of the commons ○ Both an economic and ecological concept which states that if many people have free, unlimited access to a finite, valuable resource, they will tend to overexploit it and either permanently or temporarily reduce it (IASC). ● Commons ○ A finite, valuable natural resource, shared by a community (IASC). ○ In the case of ToC, these are the stations with fish. ● Scarcity ○ When the demand for a good or service is greater than the availability fo that good or service (National Geographic). ○ As the game of ToC progresses and players continue fishing, the fish population will decrease such that fish become more scarce. ● Supply and demand ○ A basic economic concept stating the linear relationship between price and quantity supplied and the inverse relationship between price and quantity demanded. ○ In ToC, this relates to how decreased supply (ie scarcity of the fish) and/or increased demand (from the players who want to buy items from the store) makes the fish a more metaphorically "expensive" resource.

Modeling	There are several types of graphs in ToC: ● The “Fish Graph”  ○ Displays and tracks all players’ numbers of fish, both current and total ● The “Population Graph”  ○ Displays and tracks the station populations over seasons ● The “Leaderboard - ranked by Fish Left”  	Rank	Name	Fish Left
1	Playerone	88
2	Playertwo	36 | ○ Displays the students with the top numbers of fish, based on their current amount of remaining fish ● The “Leaderboard - ranked by Fish Caught”  | Rank	Name	Fish Caught
1	Playertwo	261
2	Playerone	183 | ○ Displays the students with the top numbers of fish, based on their total amount of fish caught
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	While students should not see the teacher dashboards, teachers can easily screenshot the graphs or download images of them (by pressing on the three horizontal lines in the top right corner) to show to students. The graphs can be used for a wide variety of purposes. For example: ● The population graph can demonstrate to students the decline in the number of fish at each station due to overfishing. ○ The population graph of subsequent playthroughs can be compared to the graph from the original game to serve as quantitative data to support the results of players' experimenting with strategy. I.e., if the decline was less steep or if there was no decline, that would prove their strategy worked. ● The leaderboard can be used to determine which players gained the most fish, and facilitators may want to call on those players during the class discussion. ● Similarly, the fish graph shows the same information as the leaderboard (fish caught and fish left) but for all players. This could be monitored during the game to determine when most players have caught a certain number of fish, or it can be examined afterwards to see how many fish students caught on average.
Education Standards	● ToC can easily be used to fulfill the following educational standards ○ Interdependent Relationships in Ecosystems: ■ HS-LS2-7. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity. ■ HS-LS4-6. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity. ■ HS-ESS3-4. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. ○ Engineering Design: ■ 3-5-ETS1-1. Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. ■ 3-5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem. ■ 3-5-ETS1-3. Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved. ○ See all education standards at https://www.nextgenscience.org/search-standards.

Reflection

After the class is done playing ToC, they should reflect on what they've experienced.

Common Reflection Prompts	● What was the game about? ● What was the objective? Is the game a realistic model/simulation? ● How might we limit overfishing in the real world? ● What are the ethical implications of your ideas? ● What did we learn about population biology/disheries management/sustainability? ● How did we work together to answer questions? ● What questions do you still have?

The purpose of the reflection is to help players learn that systems are made up of many parts, and problems are solved by many people, each with different perspectives. Additionally:

- Each player's participation in important – both in playing the game and in discovering how it works.
- The concepts this explores are important and connect to disciplines beyond ecology and population biology.